

task-1

Laptop Won't Power On – Troubleshooting Steps

Step	Troubleshooting Action	What You Are Checking For
1. Check the power adapter and outlet	Plug the charger into a known-working wall outlet and check whether the charging LED turns on.	Checks whether the wall outlet, charger, or power cable is supplying power.
2. Inspect the charging port	Examine the laptop's DC-in/USB-C charging port for looseness, dust, or physical damage.	Checks whether the laptop is actually receiving power from the charger .
3. Perform a power reset	Disconnect the charger and, if possible, remove the battery. Hold the power button for about 15–30 seconds , then reconnect the charger and try powering on.	Clears residual electrical charge and can resolve a temporary power-management or firmware state .
4. Check the battery	If the laptop works with the charger connected but shuts off when unplugged, test the battery/charging status. If possible, test with a known-good compatible battery or charger.	Determines whether the battery is completely discharged, defective, or not charging .
5. Check for hardware/POST signs	Try powering on and observe LEDs, fan movement, beep codes, or display activity. If there is no response, inspect internal power connections or have the motherboard/power circuit professionally tested.	Helps determine whether the problem is caused by the motherboard, power circuit, RAM, display, or another internal component .

Task-2

Helpdesk Troubleshooting Flow – Monitor Blank Screen

Problem: Monitor shows a blank/black screen, but the PC appears to be running.

Step-by-Step Troubleshooting Flow

1. Check the monitor power

- Confirm the monitor is switched on.
- Check the power cable and power LED.

- Try a different wall socket if necessary.
- **If the monitor has no power** → fix the power connection or test with a known-good cable.



2. Check the video cable

- Check the **HDMI, DisplayPort, DVI, or VGA** cable at both ends.
- Disconnect and reconnect it firmly.
- If available, test with another known-good cable.
- **If the cable is faulty** → replace it.



3. Check the correct input source

- Open the monitor's menu and make sure it is set to the correct input, such as **HDMI 1, HDMI 2, or DisplayPort**.
- **If the wrong input is selected** → select the port being used by the PC.



4. Check the PC's display connection

- If a dedicated **GPU** is installed, make sure the monitor cable is connected to the **GPU video output**, not the motherboard's video output.
- Ensure the GPU is firmly seated and its power connectors are connected.



5. Check for POST/boot problems

- Restart the PC and listen for **beep codes**.
- Check whether keyboard LEDs, fans, or other indicators behave normally.
- If there is a memory-related beep code, reseal/test the RAM.



6. Test another monitor/display

- Connect the PC to another known-working monitor or TV.
- Alternatively, connect the original monitor to another computer.
- **If another monitor works** → the original monitor or its cable/input may be faulty.

- **If no monitor works** → investigate the GPU, RAM, motherboard, or power supply.



7. Check Windows display settings

If Windows starts but the screen remains blank:

- Press **Win + P** and select **PC screen only** or **Duplicate**.
- Try **Win + Ctrl + Shift + B** to reset the graphics driver.
- Check the display resolution and refresh rate after getting a picture.



8. Confirm the fix

- Restart the PC.
- Verify that the monitor displays the POST screen and Windows normally.
- Test the system for several minutes to confirm the blank-screen problem does not return.

Quick Flow

Blank screen → Monitor power → Video cable → Correct input → GPU connection → POST/RAM check → Test another monitor → Display settings → Confirm fix

Likely causes: loose/faulty video cable, incorrect monitor input, incorrect GPU connection, graphics-driver problem, faulty GPU/RAM, or a defective monitor.

Task-3

Wireless Mouse Troubleshooting

Since the **keyboard works normally**, I would isolate the problem by testing the mouse, USB connection, and system settings separately.

Step-by-Step Diagnosis

Step	Action	What it Checks
1. Check the mouse power	Replace the mouse battery or recharge it. Check whether the mouse power/LED indicator works.	Determines whether the mouse is simply out of power .

Step	Action	What it Checks
2. Check the wireless receiver	Remove and reconnect the mouse's USB receiver. Make sure it is firmly connected.	Checks for a loose or improperly connected receiver .
3. Test another USB port	Plug the receiver into a different USB port, preferably a port directly on the PC.	Determines whether the original USB port is faulty.
4. Test the mouse on another computer	Connect the wireless receiver to another working PC and test the mouse.	If it fails there too, the mouse/receiver is likely faulty . If it works, the original PC is the likely source.
5. Check Device Manager	Open Device Manager → Mice and other pointing devices and check for warning symbols.	Checks whether Windows is detecting the mouse and its driver correctly .
6. Check system settings	Verify that the mouse is enabled and check Bluetooth settings if it is a Bluetooth mouse. Restart the PC if necessary.	Determines whether a Windows setting, driver, or Bluetooth connection is preventing operation.
7. Reinstall/update the driver	In Device Manager, uninstall the mouse device/driver and restart Windows so it can detect it again.	Resolves possible driver corruption or configuration problems .
8. Confirm the result	Test mouse movement and buttons after each correction.	Confirms whether the problem has been resolved .

How to Isolate the Fault

- **Mouse doesn't work on another PC:** likely **mouse/receiver/battery problem**.
- **Mouse works on another PC but not in the original USB port:** likely **USB-port problem**.
- **Mouse works in another USB port on the same PC:** original **USB port is the problem**.
- **Mouse works in all USB ports but not normally in Windows:** investigate **drivers and system settings**.

Task-4

Gaming PC Overheating – Hardware Troubleshooting

If a gaming-café PC overheats and shuts down during gaming, I would check these **3 hardware causes first**:

Hardware Cause	How I Would Check It
1. Faulty or blocked cooling fans	Open the case and check whether the CPU, GPU, and case fans are spinning correctly. Look for a fan that is stopped, noisy, or spinning slowly. Replace a faulty fan and ensure proper airflow through the case.
2. Dust buildup	Inspect the CPU heatsink, GPU heatsink, case fans, and air vents for dust. Heavy dust can block airflow and reduce cooling. Power off the PC and clean the components using appropriate compressed air.
3. Dried or poorly applied thermal paste	Check the contact between the CPU and its heatsink. Old or dried thermal paste can prevent efficient heat transfer. If necessary, remove the heatsink, clean the old paste, apply new thermal paste correctly, and reinstall the heatsink firmly.

Final Check

After making the repairs:

1. Start the PC and monitor CPU/GPU temperatures.
2. Run a gaming workload or suitable stress test.
3. Check that temperatures remain within the component manufacturer's normal operating range.
4. Confirm that the PC no longer shuts down unexpectedly.